PLASMA ARC EXTRACTION

High-Temperature Green Extraction Technology for Efficient Metal Recovery

WHAT IS IT?

Plasma Arc Extraction is an advanced metallurgical process that uses a plasma torch to generate extremely high temperatures (5,000–20,000 °C) to melt ore quickly and separate valuable metals from impurities.

KEY ADVANTAGES

- Very high temperature & efficiency
- Can treat low grade ores
- Fast melting and processing
- Lower emissions
- Useful for recycling e-waste
- Multi-metal recovery possible



TYPICAL ELEMENTS EXTRACTED

Fe Iron	**Cu** Copper	**Ni** Nickel	**Cr** Chromium	**Ti** Titanium
Au Gold	**Ag** Silver	**W** Tungsten	**Mo** Molybdenum	**REE** Rare Earth Elements

PROCESS FLOW



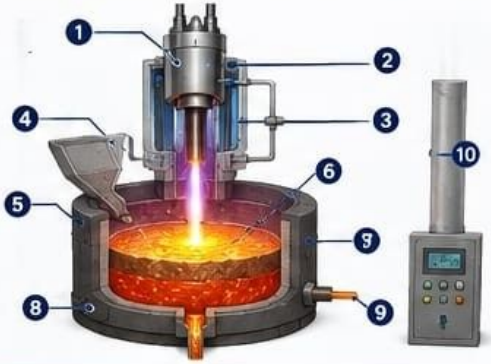
HOW IT WORKS (SCRIPT)

1. Crushed ore is fed into the furnace through the feed hopper.
2. Plasma torch generates a high-temperature plasma jet.
3. The plasma melts the ore instantly.
4. Metals and slag separate due to difference in density.
5. Slag (impurities) floats on top and is removed.
6. Molten metal collects at the bottom and is tapped.
7. Metal is further refined if high purity is required.



MAIN COMPONENTS

1. Plasma Torch
2. Power Supply (DC)
3. Plasma Gas System
4. Feed Hopper
5. Furnace Chamber
6. Electrodes
7. Cooling System
8. Slag Outlet
9. Metal Tap Hole
10. Gas Exhaust System
11. Control Panel



PLASMA GASES USED

- Argon (Ar) – Most common
- Nitrogen (N_2) – Economical
- Helium (He) – High energy
- Air / Oxygen – In some cases
- Hydrogen (H_2) – Special applications



TYPICAL TEMPERATURE

Plasma Temperature:
5,000 – 20,000 °C

Furnace Operating Temperature:
1,500 – 3,500 °C

APPLICATIONS

- Steel & metal industries
- Extraction of high melting point metals
- E-waste recycling
- Hazardous waste treatment
- Aerospace & defense metals

LIMITATIONS

- High electricity consumption
- Expensive equipment
- Electrode wear
- Skilled operation required
- Not suitable for all ores

Towards a Sustainable Future – Recover More, Waste Less, Preserve Earth.